📚 AI-Powered Educational Content Generation for West Africa & Nigeria

EnumVerse AI — System Architecture

CLIENT LAYER

🖥️ Backend Engineers
curl / SDK → REST API

📋 Educators / Stakeholders
Gradio UI on HF Spaces

🤗 Gradio 4.43 · HuggingFace Spaces (t4-medium · 16 GB VRAM)
5 tabs: Content · Curriculum · Assessment · Diagnostics · System Status

FASTAPI REST LAYER · /api/v1

GET /health	GET /skills	POST /curriculum/generate	POST /content/generate	POST /assessment/generate	POST /insights/analyze
Liveness check	Skill library browser	Pillar 1 · Schemes of Work	Pillar 2 · Lessons / Stories	Pillar 3 · WAEC-style exams	Pillar 4 · Diagnostics & Quality

LANGGRAPH ORCHESTRATION · StateGraph (LangChain + LangGraph)

route_by_pillar (Entry node) → **curriculum_node** (Pillar 1) | **generate_node** (Pillar 2) | **assessment_node** (Pillar 3) | **insights_node** (Pillar 4 → END)

skill_tag_node
Keyword match → SkillTag[]

retrieve_node
RAG retrieval with metadata filter · feeds Pillars 1-3

VALIDATION PIPELINE

rules_node
8 rules · ~50ms
No LLM

PASS → END
FAIL → **judge_node**
Mistral 7B · 5 dims
score ≥ 3.5 → END

<3.5 → **revise_node**
Guided revision
max 2 retries → rules_node

EnumVerseState (TypedDict)
pillar · request · rag_context · rag_metadata · generated_content
structured_output · skill_tags · validation_report · revision_instruction
revision_count (max 2) · final_content · error

CORE SERVICES

🧠 LLM Service (llm.py)

Primary: LLaMA 3.2 3B Instruct
meta-llama/Llama-3.2-3B-Instruct
4-bit BnB · bfloat16 · device_map=auto

Judge: Mistral 7B Instruct v0.3
mistralai/Mistral-7B-Instruct-v0.3
Triggered only on rule failure

🔍 RAG Pipeline

Embeddings: multilingual-e5-large
intfloat/multilingual-e5-large · 1024-dim
Supports Hausa / Yoruba / Igbo

Vector Store: ChromaDB (SQLite)
Pre-filter: board + subject + level
→ Qdrant at >100K docs

🏷️ Enum Skill Library

4 Domains · 119 Skills
Technical · Behavioral · Cognitive · Language
Hierarchical parent→child structure

match_from_text() auto-tagging
Keyword aliases + regex matching
Per-question + per-response SkillTag[]

📝 Prompt Engine (prompts.py)

build_curriculum_prompt()
build_content_prompt()
build_assessment_prompt()
build_judge_prompt()
Language map: en/yo/ha/ig/pcm/en-yo…
Cultural anchors always injected

STORAGE & DATA SOURCES

🗄️ ChromaDB	📄 skill_library.json	📚 Curriculum Docs	🤗 HF Datasets	🤖 Synthetic Bootstrap
data/vectorstore/ (SQLite)	data/skills/ · in-memory on load	data/raw/ · PDF / DOCX	masakhane/mafand-mt	LLaMA-generated NERDC outlines
Persistent · idempotent ingestion	Singleton via get_skill_library()	NERDC · WAEC · OER Africa	castorini/mr-tydi · opus-100	Tagged synthetic:true in metadata

TECH STACK & DESIGN NOTES

FastAPI 0.115 · Pydantic v2 · LangGraph 0.2 · ChromaDB 0.5 · Gradio 4.43 · Python 3.11+
Validation: Rules every response (~50ms) → Mistral judge only on rule failure → revise_node (max 2 retries)
Cultural anchors: Chukwuemeka · Adaeze · Aminu · Tunde · Kofi · Ama · Fatima · Balogun Market · Niger River · ₦ Naira / ₵ Cedis · NNPC · WAEC · UBEC

Chunking: Hierarchical (UNIT/TOPIC/WEEK headers) · 400-600 tokens · 80-token overlap · parent h
Languages: English · Yoruba (tonal diacritics) · Igbo (Onwu) · Hausa · Nigerian Pidgin · Twi · Ewe · M
Evaluation: ROUGE-L · BERTScore · Bloom Accuracy · Cultural Name Ratio · RAG Faithfulness · Sk